

PROPOSAL FOR DEGREE PROJECT

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1. INTRODUCTION

1.1 Context and Background

In recent years, remote work has become a widely adopted practice, especially in the technology sector. Software developers and similar professionals have experienced a significant shift in their work environment, transitioning from traditional office-based settings to remote or hybrid work models. This shift has brought multiple benefits, such as increased flexibility and reduced commuting times. However, it has also introduced new challenges, particularly regarding workers' well-being and perceived stress levels.

As the demand for remote work continues to grow, understanding its impact on employees has become a crucial concern for companies. While many studies have analyzed remote work's effects on productivity and performance, fewer have focused on its influence on well-being and stress. Identifying the factors that contribute to either a positive or negative experience in a remote setting is essential for organizations aiming to optimize working conditions and improve employee satisfaction.

1.2 Problem

Despite the increasing adoption of remote work, it remains unclear which factors influence the most the software developers' well-being and stress levels. Various elements, such as demographic characteristics, job-related factors, and work-life balance may contribute to employees' experiences in a remote setting. However, there is limited quantitative research that systematically examines these variables using robust analytical models.

1.3 Research Question

This study seeks to answer the following research question:

What are the factors that may be influencing (positively or negatively) the well-being or perceived stress level of software developers or similar workers after implementing hybrid/remote work?

This question encompasses both demographic variables and job-related factors, aiming to provide a comprehensive understanding of the determinants of well-being and stress in a hybrid/remote work environment.

1.4. Purpose of the Study

The primary objective of this research is to identify and analyze the key variables that impact software developers' well-being and stress levels in Hybrid/remote work settings. By leveraging a Partial Least Squares Structural Equation Modeling (PLS-SEM) approach with instrumental variables, this study will offer a different quantitative perspective that has been absent from previous research in this topic. The findings will enable organizations to make data-driven decisions to improve employee working conditions and better adapt to the evolving nature of remote work. This study will focus on software developers and similar technical professionals working remotely at least one day a week. It will examine both individual and work-related factors.

Lastly, this study aims to make three key contributions:

- It will apply PLS-SEM with instrumental variables, a quantitative approach that has been underutilized in previous well-being studies related to remote work.
- It will shift the focus from traditional performance-based analyses to an understanding of employee well-being and stress levels, offering a different perspective on the impact of remote work.
- It will analyze whether employees from different countries experience remote work differently, with a particular focus on Colombian employees.

2. RELATED LITERATURE

2.1 Theoretical Background

The shift to remote work has changed workplace dynamics, affecting employee well-being and stress levels. Various psychological theories help explain how these factors interact in a remote work setting, particularly for software developers. This section reviews two relevant theoretical perspectives:

The Self-Determination Theory, developed by Ryan and Deci, identifies three fundamental human needs that drive motivation and well-being: autonomy, competence, and relatedness [1]. In remote work, autonomy is often enhanced as employees have greater control over their schedules. However, the lack of social interaction can negatively impact relatedness, potentially increasing stress and reducing well-being. A study on software engineers working remotely found that competence and autonomy were positively correlated with job satisfaction, but lower relatedness led to feelings of isolation [2].

A second relevant approach is the hedonic and eudaimonic framework. This perspective distinguishes between two aspects of well-being: hedonic well-being, which relates to short-term pleasure and stress avoidance, and eudaimonic well-being, which is associated with meaning, purpose, and long-term fulfillment [3]. Research on software professionals has found that intrinsic motivators such as personal growth, mastery, and knowledge-sharing contribute

to eudaimonic well-being, while stressors such as lack of communication, isolation, and job insecurity reduce overall well-being [3].

These theoretical models offer complementary perspectives on how remote work impacts well-being and stress. Understanding these frameworks allows us to design an empirical study that examines which factors most significantly influence software developers' well-being and stress in remote work environments.

2.2 Studies on Remote work and well-being for developers

Empirical research has explored how remote work impacts employee well-being and productivity. A study by Vizcaíno et al. investigated the psychological effects of remote work on software engineers, finding that stress and motivation significantly influence performance [3]. The study used Partial Least Squares Structural Equation Modeling (PLS-SEM) to establish relationships between these factors and revealed that communication and trust among team members are critical in reducing stress levels.

Another study by Santos et al. focused on hybrid work models and their impact on developers' satisfaction. The study concluded that hybrid models allow employees to maintain work-life balance while also benefiting from in-person collaboration when needed. However, the research also identified challenges such as feelings of isolation and difficulty in setting clear work boundaries, which contribute to stress [12].

Lastly, different methodological approaches have been used to study well-being and stress in remote work environments:

- **Survey-Based Research:** Multiple surveys have been conducted to measure employee perceptions of stress, well-being, and productivity, with the results primarily analyzed with descriptive methods.
- **Partial Least Squares Structural Equation Modeling:** PLS-SEM has been increasingly used to analyze the causal relationships between well-being factors, demonstrating their suitability for quantitative analysis in the developers' environments [4].

Despite the extensive research on remote work, several key gaps remain in the current literature:

Lack of Mathematical Modeling in Well-Being Studies: Most studies on remote work rely on descriptive analysis rather than applying advanced mathematical models. While surveys provide valuable insights into employee perceptions, they often lack quantitative rigor to establish strong causal relationships between stress, well-being, and work modalities factors.

Limited Use of Advanced Statistical Methods: Among the few studies that employ mathematical models [3][4], the predominant approaches are Structural Equation Modeling (SEM) or Partial Least Squares SEM (PLS-SEM). However, there is a lack of research utilizing PLS-SEM with instrumental variables, which could enhance causal inference and provide a more robust understanding of the factors influencing stress and well-being.

Focus on Performance Over Well-Being or Stress perceptions: Most existing studies focused on employee performance and productivity rather than focusing on well-being and stress perception. While performance metrics are essential, understanding how remote work affects current employee's well-being and long-term job satisfaction is equally critical for organizations adapting to new work models.

This literature review provides a strong foundation for understanding how remote work influences well-being and stress. However, addressing the mentioned gaps will contribute to a more comprehensive understanding of this new environment work dynamics.

3. OBJECTIVES

The primary goal of this research is to analyze the factors influencing the well-being and perceived stress levels of software developers and similar professionals working hybrid/remotely. This study aims to identify key demographic and job-related variables that contribute to these experiences, providing insights that can help organizations improve working conditions and adapt to the evolving landscape of remote work.

3.1 General Objective

- Determine the factors that positively or negatively influence the well-being and perceived stress levels of software developers and similar professionals in a remote work environment, using a PLS-SEM approach with instrumental variables.

3.2 Specific Objectives

- Identify demographic factors (e.g., age, gender, education level) that may impact well-being and stress levels in remote software developers.
- Analyze job-related variables (e.g., job responsibilities, career growth opportunities, motivation) and their relationship to perceived stress and well-being.
- Assess the influence of work-life balance on stress levels and overall well-being in remote work environments.
- Analyze whether employees from different countries experience remote work differently, with a particular focus on Colombian employees.

- Apply a quantitative methodology (PLS-SEM with instrumental variables) to systematically measure and evaluate the relationships between these factors.
- Compare findings with existing literature to determine whether the results align with or challenge previous studies.

This study will contribute to the existing knowledge by changing the focus from productivity-based evaluations of remote work to an analysis centered on employee well-being and stress perception. By leveraging PLS-SEM with instrumental variables, it will introduce a more robust and data-driven approach to understanding the challenges and benefits of remote work in the tech industry.

4. METHODOLOGY

This study will employ a quantitative approach to analyze the factors influencing the well-being and perceived stress levels of software developers and similar professionals in a hybrid/remote work environment. A structured survey was conducted across four countries: Argentina, Colombia, Spain, and Mexico. The survey gathered data on participant's perceptions of stress levels, well-being, and productivity, comparing their experiences in remote and in-person work modalities. The collected data will be analyzed using a Partial Least Squares Structural Equation Modeling (PLS-SEM) approach with instrumental variables to examine the relationships between key factors.

4.1 Research Design and Survey Data Collection

This research will follow a cross-sectional survey design, using the collected responses from professionals who have worked in both in-person and remote environments for at least two years, this survey was designed and collected on a previous study from Vizcaino and Garces [16]. The study will seek to identify key demographic and job-related factors that impact well-being and stress levels.

In the previous study [16], a structured online survey was developed and deployed, consisting of 100 questions. The survey was designed to capture participants' perceptions of stress levels, well-being, and productivity, comparing their experiences in both work modalities. The survey included the following specifications:

- **Target Population:** Software developers, engineers, managers and other technical professionals working in the software industry with experience in both remote and in-person work environments.
- **Inclusion Criteria:** At least two years of work experience in in-person modality and currently working hybrid or remote.

- **Geographical Scope:** The survey was deployed in Argentina, Colombia, Spain, and Mexico to provide cross-cultural insights.

4.2 Variables and Measurement

The study will examine a range of demographic, job-related, and psychosocial factors that may influence well-being and stress levels.

Independent Variables:

- **Demographic factors:** Age, gender, education level, work experience.
- **Job-related factors:** Job role, career growth opportunities, workload, company support, motivation.
- **Work-life balance factors:** Flexibility, autonomy, social isolation, time management.

Dependent Variables:

- Well-being perception score
- Perceived stress levels
- Instrumental Variables: Team collaboration dynamics, company policies, among others.

4.3 Data Analysis Approach

PLS-SEM (Partial Least Squares Structural Equation Modeling): Will be used to examine complex relationships between demographic, job-related, and psychosocial variables. This methodology will allow for latent variable modeling and the handling of non-normal data distributions.

Instrumental Variables Approach: Will help address potential endogeneity issues in the model and will improve the reliability of causal inferences.

5. EXPECTED RESULTS

Based on prior research and the collected data, the following outcomes are expected:

Demographic and job-related factors will significantly influence well-being and stress levels:

- Age, gender, nationality and education level may show variations in stress perception.
- Career growth opportunities and company support are expected to correlate with higher well-being scores.

Remote work will present both advantages and challenges:

- Increased flexibility and autonomy are likely to positively impact well-being.
- Social isolation and difficulties in work-life balance may contribute to higher stress levels.

Inconsistencies in responses in terms of perceptions:

- Some professionals may report higher productivity, stress level or well-being in remote settings, while others may differ.
- Factors such as self-discipline, company policies, and collaboration tools may influence perceived productivity differences.

PLS-SEM analysis will reveal key interactions between stress, well-being, and productivity.

- It is expected that work-life balance and support in job activities will be key in the relationship between remote work and well-being.
- Instrumental variables analysis will provide additional insights into underlying causal mechanisms.

Findings will help shape better remote work policies.

- Companies will gain a data-driven understanding of employee needs in remote environments.
- The study will offer recommendations to improve well-being, reduce stress, and optimize remote work conditions.

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